GLUEING SCHEMES OVER A BASIS FROM COMPATIBLE LOCAL DATA

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Editorial note. One '(easy) verification' omitted: the constructed system (g, {h_U}) lies in F_W(T) and maps to alpha under the displayed arrow; the construction of the h_U themselves is fully displayed via conditions (a) and (b).

Allowed standard prerequisites (editorial).

- Glue-functors criterion. schemes-lemma-glue-functors (cross-chapter, named). Provides the four-step representability template.
- Glueing of sheaves. sheaves-section-glueing-sheaves (cross-chapter, named). Underlies the sheaf property of the constructed functor.

Source reference. Original reference 01JJ: focused-seeds/geometry/S9-cons-relative-glueing/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{B}$ be a basis for the topology of S. Suppose given the following data:*

(1) *For every $U \in \mathcal{B}$ a scheme $f_U : X_U \to U$ over U.*
(2) *For $U, V \in \mathcal{B}$ with $V \subset U$ a morphism $\rho^U_V : X_V \to X_U$ over U.*

Assume that

(a) *each ρ^U_V induces an isomorphism $X_V \to f_U^{-1}(V)$ of schemes over V,*
(b) *whenever $W, V, U \in \mathcal{B}$, with $W \subset V \subset U$ we have $\rho^U_W = \rho^U_V \circ \rho^V_W$.*

Then there exists a morphism $f : X \to S$ of schemes and isomorphisms $i_U : f^{-1}(U) \to X_U$ over $U \in \mathcal{B}$ such that for $V, U \in \mathcal{B}$ with $V \subset U$ the composition

$$X_V \xrightarrow{i_V^{-1}} f^{-1}(V) \xrightarrow{inclusion} f^{-1}(U) \xrightarrow{i_U} X_U$$

is the morphism ρ^U_V. Moreover X is unique up to unique isomorphism over S.

Proof. To prove this we will use Schemes, Lemma 01JJ. First we define a contravariant functor F from the category of schemes to the category of sets. Namely, for a scheme T we set

$$F(T) = \left\{ \begin{array}{c} (g, \{h_U\}_{U \in \mathcal{B}}),\ g : T \to S,\ h_U : g^{-1}(U) \to X_U, \\ f_U \circ h_U = g|_{g^{-1}(U)},\ h_U|_{g^{-1}(V)} = \rho^U_V \circ h_V \ \forall\ V, U \in \mathcal{B}, V \subset U \end{array} \right\}.$$

The restriction mapping $F(T) \to F(T')$ given a morphism $T' \to T$ is just gotten by composition. For any $W \in \mathcal{B}$ we consider the subfunctor $F_W \subset F$ consisting of those systems $(g, \{h_U\})$ such that $g(T) \subset W$.

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First we show F satisfies the sheaf property for the Zariski topology. Suppose that T is a scheme, $T = \bigcup V_i$ is an open covering, and $\xi_i \in F(V_i)$ is an element such that $\xi_i|_{V_i \cap V_j} = \xi_j|_{V_i \cap V_j}$. Say $\xi_i = (g_i, \{h_{i,U}\})$. Then we immediately see that the morphisms g_i glue to a unique global morphism $g : T \to S$. Moreover, it is clear that $g^{-1}(U) = \bigcup g_i^{-1}(U)$. Hence the morphisms $h_{i,U} : g_i^{-1}(U) \to X_U$ glue to a unique morphism $h_U : g^{-1}(U) \to X_U$. It is easy to verify that the system $(g, \{h_U\})$ is an element of $F(T)$. Hence F satisfies the sheaf property for the Zariski topology.

Next we verify that each F_W, $W \in \mathcal{B}$ is representable. Namely, we claim that the transformation of functors

$$F_W \longrightarrow \mathrm{Mor}(-, X_W),\ (g, \{h_U\}) \longmapsto h_W$$

is an isomorphism. To see this suppose that T is a scheme and $\alpha : T \to X_W$ is a morphism. Set $g = f_W \circ \alpha$. For any $U \in \mathcal{B}$ such that $U \subset W$ we can define $h_U : g^{-1}(U) \to X_U$ be the composition $(\rho_U^W)^{-1} \circ \alpha|_{g^{-1}(U)}$. This works because the image $\alpha(g^{-1}(U))$ is contained in $f_W^{-1}(U)$ and condition (a) of the lemma. It is clear that $f_U \circ h_U = g|_{g^{-1}(U)}$ for such a U. Moreover, if also $V \in \mathcal{B}$ and $V \subset U \subset W$, then $\rho_V^U \circ h_V = h_U|_{g^{-1}(V)}$ by property (b) of the lemma. We still have to define h_U for an arbitrary element $U \in \mathcal{B}$. Since $\mathcal{B}$ is a basis for the topology on S we can find an open covering $U \cap W = \bigcup U_i$ with $U_i \in \mathcal{B}$. Since g maps into W we have $g^{-1}(U) = g^{-1}(U \cap W) = \bigcup g^{-1}(U_i)$. Consider the morphisms $h_i = \rho_{U_i}^U \circ h_{U_i} : g^{-1}(U_i) \to X_U$. It is a simple matter to use condition (b) of the lemma to prove that $h_i|_{g^{-1}(U_i) \cap g^{-1}(U_j)} = h_j|_{g^{-1}(U_i) \cap g^{-1}(U_j)}$. Hence these morphisms glue to give the desired morphism $h_U : g^{-1}(U) \to X_U$. We omit the (easy) verification that the system $(g, \{h_U\})$ is an element of $F_W(T)$ which maps to α under the displayed arrow above.

Next, we verify each $F_W \subset F$ is representable by open immersions. This is clear from the definitions.

Finally we have to verify the collection $(F_W)_{W \in \mathcal{B}}$ covers F. This is clear by construction and the fact that $\mathcal{B}$ is a basis for the topology of S.

Let X be a scheme representing the functor F. Let $(f, \{i_U\}) \in F(X)$ be a "universal family". Since each F_W is representable by X_W (via the morphism of functors displayed above) we see that $i_W : f^{-1}(W) \to X_W$ is an isomorphism as desired. The lemma is proved. $\square$